Network Traffic Analysis, Nmap and Windows Logging

Intern name	**Kunal Jain**
Internship / organisation	DG Interns Hub
Week	Week 2
Reporting period	24 - 30 October 2026

1. Executive Summary

Week 2 focused on understanding how network activity appears in packet captures and how Windows records security-relevant activity in event logs. The learning objectives covered TCP connection behaviour, common network protocols, Wireshark filtering, Nmap scan techniques, Windows components, Security Event IDs, and logon types. These concepts support later endpoint monitoring and intrusion detection work.

The practical plan used an Ubuntu and a Kali Linux virtual machine for traffic generation and scanning, a Windows virtual machine for event generation and log analysis, Wireshark for packet inspection, Nmap for host and service discovery, and PowerShell for event collection. TryHackMe rooms and official Wireshark sample captures provided additional guided practice.

2. Objectives and Scope

- Explain the TCP three-way handshake, connection teardown, and the meaning of the main TCP flags.
- Recognise normal and potentially suspicious DNS, HTTP, TLS, ARP, ICMP, and SMB patterns.
- Use Wireshark capture filters and display filters appropriately.
- Compare Nmap discovery, TCP, UDP, version-detection, OS-detection, and aggressive scan behaviour.
- Understand Windows processes, services, registry, users and groups, UAC, and Event Viewer logs.
- Interpret selected Windows Security Event IDs and logon types.
- Establish why baselines are important to security operations centre (SOC) analysis.

3. Technical Learning

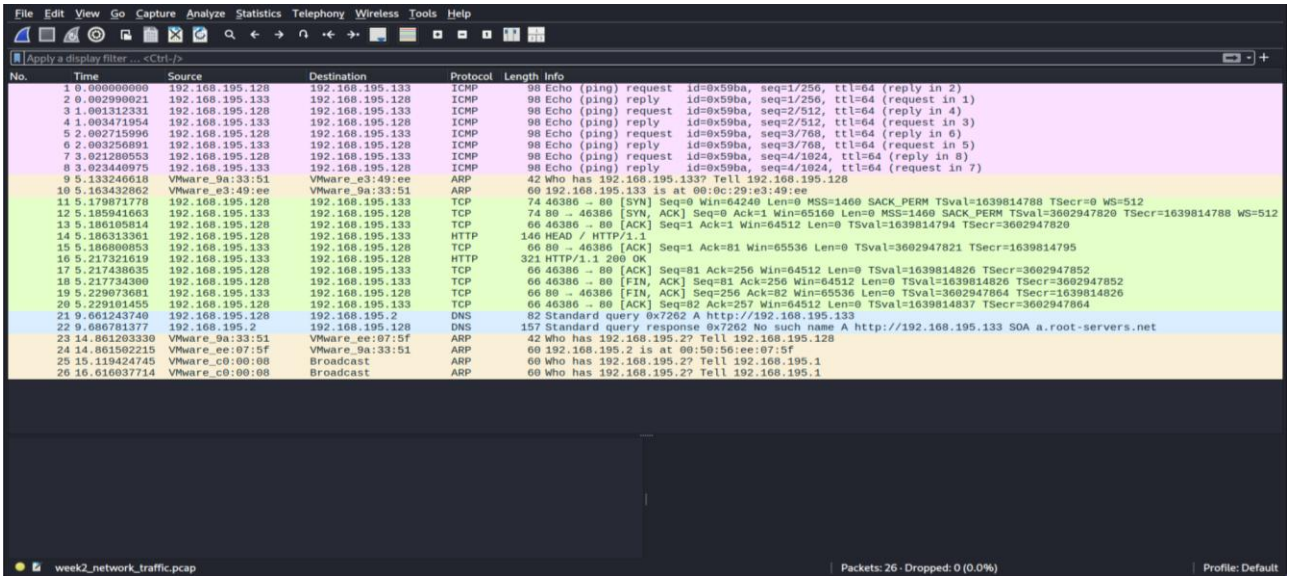
3.1 TCP handshake and teardown

TCP normally establishes a connection using SYN → SYN/ACK → ACK. The client sends SYN to request a connection; the server replies with SYN/ACK; the client returns ACK. Data can then flow. A normal graceful close commonly uses FIN and ACK exchanges in both directions. RST indicates an abrupt reset rather than a graceful close.

Flag	Meaning	Analyst interpretation
SYN	Synchronise sequence numbers; initiate a connection.	A new connection attempt. Many SYNs to many ports/hosts can indicate scanning.
ACK	Acknowledgement field is valid.	Common in established traffic and handshake responses; interpret with connection state.
FIN	Sender has finished sending data.	Normal graceful close; unusual patterns may merit context review.
RST	Reset the connection.	Closed port response or abrupt termination; bursts can accompany scans.
PSH	Request prompt delivery of buffered data to the applica-	Often seen with application data; not inherently suspicious.
URG	Urgent pointer field is significant.	Rare in ordinary modern traffic; investigate unexpected use in context.

3.2 Protocol analysis

Protocol	Normal behaviour	Potentially suspicious behaviour
DNS	Clients query resolvers for names; responses return records such as A, AAAA or CNAME.	Unusual domains, high query volume, repeated NXDOMAIN, long/random-looking names or unexpected external resolvers.
HTTP	Readable requests and responses, expected methods, hostnames and status codes.	Unexpected methods, suspicious paths, clear-text credentials, unusual user agents or repeated errors.
HTTPS/TLS	Encrypted application data after a TLS handshake; certificate and SNI metadata may be visible depending on version and configuration.	Invalid/unexpected certificates, obsolete TLS, unusual destinations, suspicious SNI or rare cipher behaviour.
ARP	Local-network IP-to-MAC resolution and occasional gratuitous ARP announcements.	Conflicting IP/MAC mappings, unsolicited replies or rapid changes that may indicate spoofing.
ICMP	Echo requests/replies and network error messages used for diagnostics and path discovery.	Unexpected volume, sweeps, tunnelling-like patterns or traffic inconsistent with the host's role.
SMB	File/printer sharing and Windows domain-related communication within expected networks.	Unexpected SMB from internet-facing hosts, unusual shares, excessive authentication failures or lateral movement patterns.

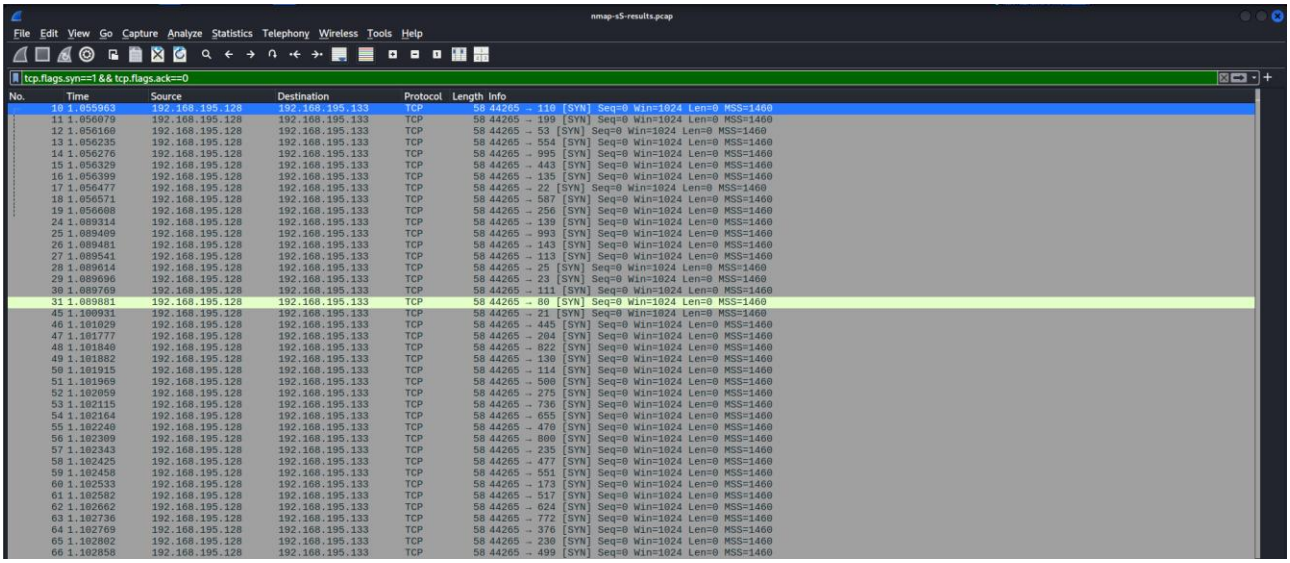


3.3 Wireshark filters

Capture filters are applied before packets are saved and use the capture-filter syntax (BPF). Display filters are applied after capture to show only matching packets; they use Wireshark's display-filter language. A capture filter can permanently exclude packets from the saved capture, while a display filter does not remove them from the file.

Type	Example	Use
Capture	host 192.0.2.10	Capture traffic to or from one host.
Capture	net 192.0.2.0/24	Limit capture to a subnet.
Capture	tcp port 80	Capture TCP port 80 traffic.
Capture	udp port 53	Capture DNS over UDP.
Capture	icmp	Capture ICMP traffic.
Display	tcp.flags.syn == 1 && tcp.flags.ack == 0	Show initial TCP SYN packets.

Display	dns	Show DNS packets.
Display	http.request	Show HTTP requests.
Display	tls.handshake	Show TLS handshake packets.
Display	ip.addr == 192.0.2.10	Show packets involving a host.



4. Nmap Scan Techniques

Option	How it works	Typical network/log evidence
-sn	Host discovery only; skips the port scan. Probes vary by privilege and network.	ARP, ICMP or TCP/UDP discovery probes; firewall and IDS logs may record host-discovery activity.
-sS	TCP SYN (half-open) scan. Sends SYN; open ports commonly reply SYN-ACK, closed ports RST; scanner usually does not complete the handshake.	Burst of SYN packets and SYN-ACK/RST responses; may appear as incomplete connections.
-sT	TCP connect scan using the operating system connect() call; completes the TCP handshake.	Full connection attempts are more likely to appear in application/service logs.
-sU	Sends UDP probes and interprets UDP responses or ICMP port- unreachable messages.	UDP probes, possible UDP responses and ICMP errors; results can be slow or ambiguous.
-sV	Probes open ports to identify service and version.	Application-level probes and replies; may create service logs or trigger IDS signatures.
-O	Uses TCP/IP stack fingerprinting to estimate the target OS.	A series of probes with differing flags, options and packet sizes; inference is not always exact.
-A	Enables OS detection, version detection, script scanning and traceroute.	Broader, more distinctive probe set; greater visibility in network, host and IDS logs.

Use only against systems for which permission has been granted. Scan results depend on privileges, firewall rules, network path, target configuration and Nmap version.

```
┌──(shivaay㉿Kali)-[~/Desktop]
└─$ sudo nmap -sn 192.168.195.133 -oN ubuntu-sn.txt -oX ubuntu-sn.xml
[sudo] password for shivaay:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 12:51 +0530
Nmap scan report for 192.168.195.133
Host is up (0.0030s latency).
MAC Address: 00:0C:29:E3:49:EE (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.65 seconds

┌──(shivaay㉿Kali)-[~/Desktop]
└─$ sudo nmap -sS 192.168.195.133 -oN ubuntu-sS.txt -oX ubuntu-sS.xml
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 12:52 +0530
Nmap scan report for 192.168.195.133
Host is up (0.0026s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
80/tcp open  http
MAC Address: 00:0C:29:E3:49:EE (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.85 seconds

┌──(shivaay㉿Kali)-[~/Desktop]
└─$ sudo nmap -sV 192.168.195.133 -oN ubuntu-sV.txt -oX ubuntu-sV.xml
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 12:53 +0530
Nmap scan report for 192.168.195.133
Host is up (0.00029s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
80/tcp open  http    Apache httpd 2.4.66 ((Ubuntu))
MAC Address: 00:0C:29:E3:49:EE (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.48 seconds
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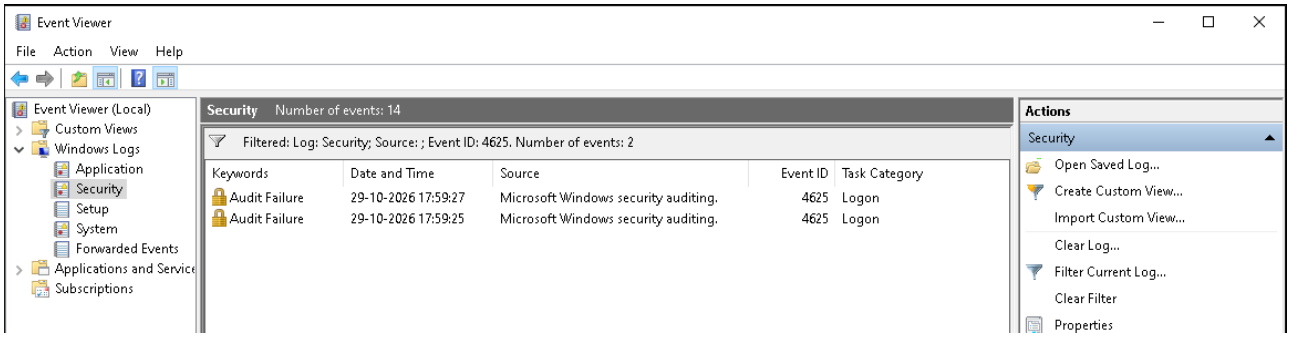
5. Windows Fundamentals and Event Logging

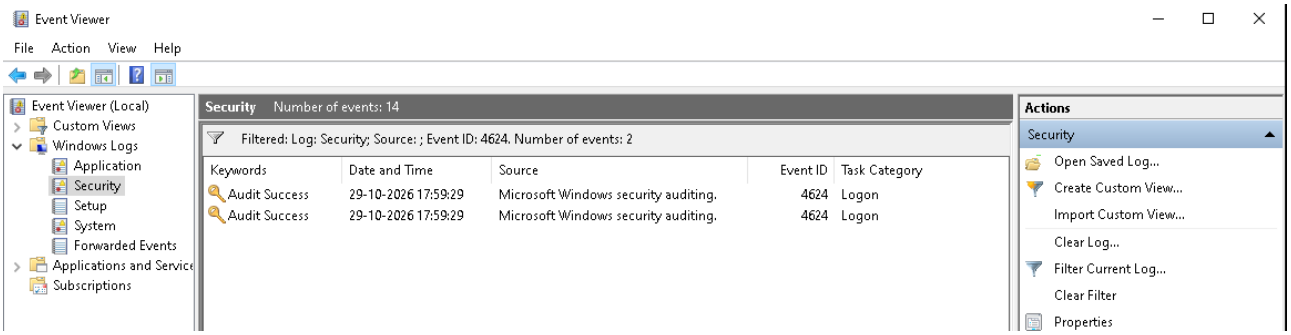
Component	Summary
Processes	Running instances of programs. Each process has a PID, security context, memory and associated threads.
Services	Background components managed by the Service Control Manager; may start automatically, manually or on demand.
Registry	Hierarchical configuration database for Windows, applications, users and system settings.
Users and groups	Security principals used for authentication and authorisation. Group membership grants sets of permissions.
UAC	User Account Control prompts or uses elevation boundaries when administrative privileges are required.
Event Viewer logs	Application records application events; Security records audited security activity; System records Windows components, drivers and services.

5.1 Key Security Event IDs

Event ID	Meaning	When an analyst uses it
4624	Successful logon.	Establish who logged on, when, from where and with which logon type.
4625	Failed logon.	Investigate password guessing, mistyped credentials and authentication failures.
4634	Logoff.	Correlate the end of a logon session.
4648	Logon attempted using explicit credentials.	Investigate runas, scheduled tasks or unexpected use of alternate credentials.

Event ID	Meaning	When an analyst uses it
4672	Special privileges assigned to a new logon.	Review privileged sessions and whether the account normally has those rights.
4688	A new process was created.	Trace process execution; command-line data may be available if auditing is configured.
4697	A service was installed.	Investigate new or unexpected services and persistence attempts.
4720	A user account was created.	Verify authorised account provisioning.
4722	A user account was enabled.	Check whether reactivation was expected.
4724	An attempt was made to reset an account password.	Review password resets, especially for privileged accounts.
4732	A member was added to a security-enabled local group.	Check additions to Administrators and other sensitive groups.
4768	A Kerberos authentication ticket-granting ticket (TGT) was requested.	Investigate Kerberos authentication patterns and failures.
4769	A Kerberos service ticket was requested.	Review access to services and unusual ticket request patterns.
4776	Credential validation using NTLM.	Investigate NTLM authentication successes/failures and source workstations.
1102	The Security audit log was cleared.	Treat unexpected clearing as high priority and investigate who performed it.





Event IDs may require specific audit policy settings. Event 4697 is a Security log event when the relevant auditing is enabled; service installation is also commonly recorded as System Event 7045. Event 1102 is generated when the Security audit log is cleared.

5.2 Windows logon types

Type	Meaning	Normal example	Suspicious example
2	Interactive	User signs in at the local keyboard.	Unexpected local sign-in to a server after hours.
3	Network	Access to a shared folder over the network.	Many failures or access from an unusual workstation.
4	Batch	Scheduled task runs under an account.	Unexpected batch logon by a personal user account.
5	Service	A Windows service starts under its configured account.	Unknown service account repeatedly logging on.
7	Unlock	User unlocks an existing session.	Repeated unlocks at unusual times or locations.
10	RemoteInteractive	User connects through Remote Desktop.	RDP logon from an unapproved external address.
11	CachedInteractive	Laptop user signs in with cached domain credentials while offline.	Cached logon on a device or account where it is unexpected.

6. Baselines and SOC Analysis

A baseline is a documented picture of normal activity for a system, user, network or environment over time. It can include usual login times, source devices, network destinations, ports, services, process activity and traffic volume.

Why it matters:

- It helps distinguish routine administration from unusual behaviour.
- It reduces false positives by adding context to alerts.
- It makes changes and deviations easier to detect.
- It supports prioritisation: a rare event on a sensitive server may deserve more attention than the same event on a lab workstation.
- Baselines must be reviewed as systems and business activity change.

7. Practical Work Completed

The following practical work was completed in the lab environment. Add exact target names, dates, commands, result summaries and evidence filenames from your own records before final submission.

Activity	Work performed / evidence to attach
Traffic capture	Installed Apache on Ubuntu; generated ping, curl and nslookup traffic from Kali while capturing in Wireshark. Evidence: TCP handshake, HTTP GET, DNS query, ICMP echo and saved PCAP.
Nmap scanning	Ran -sn, -sS, -sV, -O and -A against the authorised Ubuntu and Windows VMs. Saved normal and XML output using -oN and -oX. Captured SYN scan traffic using tcp.flags.syn==1 && tcp.flags.ack==0.
Windows event generation	Generated and located events for failed/successful logon, account creation, Administrators group membership, service creation/deletion and Security log clearing. Evidence: Event Viewer filtered-log screenshots.

PowerShell hunting	Used Get-WinEvent with FilterHashtable to export Event ID 4625 records to CSV and Group-Object to count failures by account.
TryHackMe	Worked through the listed Nmap and Windows rooms. Attach room completion screenshots or certificates where available.
Wireshark sample captures	If used as fallback/extra work, document the three selected sample captures, protocols observed, what happened and any suspicious indicators.

8. Learnings Outcome and Week 3 Readiness

By completing the activities, the intended outcomes are to be able to read a basic TCP conversation, filter packet captures efficiently, recognise scan-related packet patterns, understand what Nmap probes reveal, navigate Windows event logs, interpret common authentication and account-management events, and use PowerShell to extract and summarise security telemetry.

For Week 3, the next step is to correlate network indicators with endpoint telemetry: identify scan-like patterns, review relevant Snort alerts, and connect suspicious source/destination activity with Windows events such as 4624, 4625, 4688, 4697 and 4732.

9. Conclusion

Week 2 developed practical familiarity with packet analysis, network scanning and Windows event logging. The combination of PCAP evidence, Nmap outputs and Windows event records provides a foundation for investigating activity from both network and endpoint perspectives. The quality of the final submission depends on attaching the original screenshots, PCAP files, command outputs and room writeups referenced in this report.